

# Cryptology

## THEORY ASSIGNMENT 01

### **Security Features in Real World Applications**

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Date: 11 August 2026

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|---------|--------------------------------------------------------------------------------|
| Table 1 | <i>Security feature, mechanism and protocol for five everyday applications</i> |
| Table 2 | <i>Controls classified by CIA, mapped to prevent, detect and recover</i>       |

## 1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

We use many apps every day without thinking about the security working behind them.

Below are five common apps. For each one, we look at the main security feature it needs, how it is achieved, and the protocol used.

| Application        | Security Feature | Security Mechanism                   | Protocol / Technology            | How it keeps you safe                                                                                                                          |
|--------------------|------------------|--------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Instagram          | Authentication   | Password + login approval on phone   | 2-Factor Authentication (2FA)    | Even if someone gets your password, they still need your phone to approve the login. This stops most account takeovers.                        |
| Paytm              | Confidentiality  | Encrypted payment data + PIN         | TLS, UPI PIN                     | Your payment details are locked in a secret code while traveling over the internet, so no one snooping on the network can read them.           |
| Google Drive       | Integrity        | File checksums                       | SHA-256 hashing                  | Every file gets a unique digital fingerprint. If the file is changed even slightly, the fingerprint changes too, so tampering is easy to spot. |
| Zoom               | Authorization    | Host controls and meeting roles      | Role-based access, waiting rooms | The host decides who can join, share their screen, or unmute. Random people can't just walk into a meeting and take over.                      |
| Amazon Prime Video | Availability     | Servers spread across many locations | CDN (Content Delivery Network)   | Videos are stored on servers all over the world. If one server is busy or down, another one nearby takes over, so streaming doesn't stop.      |

Table 1: Security feature, mechanism and protocol for five everyday applications

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Every security control fits under one of three goals: Confidentiality, Integrity, or Availability (together called the CIA triad).

The table below shows how each app's controls work in three simple stages: stopping an attack (Prevent), noticing it (Detect), and fixing the damage (Recover).

| Application        | Control & Protocol           | Prevent                                                          | Detect                                                          | Recover                                                             |
|--------------------|------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------|
| Instagram          | Authentication: 2FA          | A second login step stops hackers who only have your password.   | You get an alert if someone logs in from a new device or place. | You can log out all devices and reset your password right away.     |
| Paytm              | Confidentiality: TLS + PIN   | Your PIN and encryption stop others from making payments as you. | Paytm watches for odd spending patterns and flags them.         | You can freeze your account and get wrongly charged money reversed. |
| Google Drive       | Integrity: file hashing      | Only people you allow can edit or upload files.                  | A changed file hash shows something was altered.                | Old versions of the file can be restored with one click.            |
| Zoom               | Authorization: host controls | Waiting rooms and passwords keep out uninvited guests.           | The host sees everyone who joins and can spot outsiders.        | The host can remove intruders and lock the meeting instantly.       |
| Amazon Prime Video | Availability: CDN            | Traffic is spread out so no single server gets overloaded.       | Monitoring tools catch slowdowns or server failures fast.       | Viewers are automatically switched to a working server nearby.      |

Table 2: Controls classified by CIA, mapped to prevent, detect and recover